Can we guess how much electricity Portugal will use tomorrow?

18 months of hourly data. Time series, forecasting models and a bit of fun.

13%

less electricity on the two quietest days of the week. Demand follows a weekly rhythm.

356 → 104

MW of error when guessing tomorrow: copying today vs. a model that knows the weekly rhythm.

7,773 → 7,191

AIC (lower is better) when a weekly cycle was added to the ARIMA. The missing piece was the week.

7 days ahead is a different story

The simple models (204 MW) barely beat "same weekday last week" (223 MW). Honest limits matter.